

Yang-Mills equation: The necessity of mass due to interference

[Original Problem]

"An arbitrary compact simple gauge group" G In contrast, quantum Yang-Mills theory $\mathbb{R}^4$ Located above, mass gap $\Delta > 0$ " Prove

that for any compact simple gauge group G , a quantum Yang-Mills theory exists on R^4 and has a mass gap $\Delta > 0$.

The goal is to mathematically explain why certain elementary particles (such as particles that mediate the strong interaction) have "weight (mass)," unlike photons which have zero mass.

1. Intermediate development of the mass gap: defined node

Step A: In a space free from energy diffusion and constraint

inconsistencies, energy does not simply scatter, but is distributed according to the geometric constraints (topology) of the space.

$$\text{Energy Density} \propto \text{Curvature } (F)$$

In a consistent space, energy is distributed according to geometric constraints rather than simply dissipating.

Step B: Formation of "nodes" by mathematical ratios.

As long as the ratio "1 is 0.5 of 2" is maintained mathematically, energies reinforce each other at specific points (nodes), forming the smallest unit. This is the true nature of mass.

$$\Delta F + [A, F] = \text{Mass Gap } (\Delta > 0)$$

As long as the ratio "1 is 0.5 of 2" is maintained, energy forms "nodes" through interference, inevitably resulting in a mass gap.

Step C: If a consistent cumulative vector

gauge field (cumulative vector) contains no inconsistencies, it smoothly fills space, but its interference fringes as "standing waves" define a minimum energy (mass) that is not zero.

$$\text{Mass} = \text{Minimum Interference Unit}$$

The absence of inconsistency dictates that the minimum unit of energy interference manifests as mass.

Critique of modern physics and mathematics: The reversal of reality and shadow

The fallacy of modern particle physics lies in **its view of mass solely as an "externally given adduct (such as the Higgs particle)," while failing to recognize the inherent inevitability created by the mathematical structure itself**.

- **Construction from "Nothing":** Mathematicians wonder how mass can arise from empty space, but the very existence of mathematical ratios (determinants) already promises a structure (mass) through "interference."
- **The labyrinth of calculations:** We become engrossed in the complex calculations of quantum mechanics, failing to notice the extremely simple coercive force behind it—the "absence of inconsistency"—that bundles energy into particles.
- **A misunderstanding of the true nature of mass:** Mass is not a physical quantity like the "weight" of a substance, but merely another name for a "defined density" in logical space.

The self-evident fact that "ratios are maintained in mathematics" does not allow space to be empty. The existence of a mass gap is the very "logical thickness" that makes mathematics valid as it is.

Modern physics errs by treating mass as an external additive, failing to see it as a necessary density of logic within the mathematical structure.